CASE STUDY REPORT

Gated Sparsity with L1 Regularization on CIFAR-10

Learning Compact Neural Networks via Learnable Gates & L1 Regularization

Submitted To	Tredence Analytics
Prepared By	Mrinal Seth
Email	mseth1_be23@thapar.edu
Institution	Thapar Institute of Engineering & Technology
Programme	B.E. (2023 Batch)
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Introduction

This report presents a case study on Gated Sparsity with L1 Regularization applied to the CIFAR-10 image classification benchmark. The core objective was to design a neural network capable of autonomously identifying and removing redundant weights during training, rather than relying on post-hoc pruning techniques. The mechanism used is a combination of learnable gate parameters and L1 regularization, enabling the model to learn compact representations end-to-end.

1. Problem Context

Modern deep neural networks are heavily over-parameterized. While this over-parameterization often aids convergence and generalization during training, it leads to inflated memory requirements and slower inference at deployment. The central challenge addressed in this case study is: can a network learn, during training itself, which weights are dispensable?

Rather than pruning after training (post-hoc pruning), the proposed approach integrates the pruning decision into the learning process using learnable gates and sparsity-inducing regularization. A feed-forward Multi-Layer Perceptron (MLP) was used as the backbone architecture, with each weight paired with a corresponding gate parameter that governs its contribution.

2. Approach

2.1 Gated Weight Mechanism

Each weight in the network is paired with a learnable gate parameter. During the forward pass, the effective weight contributing to computation is computed as:

```
effective_weight = weight × sigmoid(gate_score)
```

The sigmoid function constrains the gate output to the range [0, 1], providing a smooth, differentiable approximation to binary on/off switching. A gate value near 0 effectively silences the corresponding weight, while a value near 1 allows it to contribute fully. This soft gating is critical — it keeps the computation graph differentiable and enables gradient-based optimization.

2.2 L1 Regularization on Gates

To drive gate values toward zero (and thus encourage sparsity), an L1 penalty term is added to the training objective. The total training loss is defined as:

```
Loss = CrossEntropyLoss + λ × SparsityLoss
```

where SparsityLoss is the sum of absolute gate values across the network, and λ (lambda) is a hyperparameter controlling the strength of the sparsity penalty. Larger λ values push more gates toward zero, producing sparser models at the cost of potentially reduced accuracy.

2.3 Lambda Sweep

A systematic sweep over multiple values of λ was conducted to characterize the accuracy-sparsity trade-off. This allows selection of an operating point that balances model compactness with predictive performance.

2.4 Training Strategy (Important)

To stabilize training, a **warm-up phase** was used:

- First 5 epochs → only CrossEntropy loss
- Remaining epochs → CrossEntropy + sparsity loss

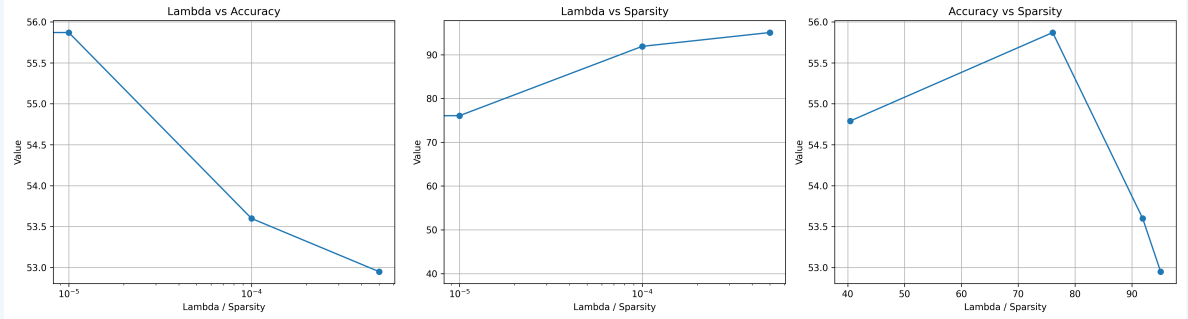
This prevents premature pruning before the model learns meaningful features.

3. Observations and Results

The experiments revealed a consistent and interpretable trade-off between sparsity and classification accuracy across all tested values of λ:

NOTEBOOK 1:-

λ (Lambda)	Train Accuracy (%)	Test Accuracy (%)	Sparsity (%)
0	55.35	55.35	0.00
1E-05	55.40	55.37	40.33
1E-04	53.31	10.61	53.34
0.01	24.59	10.00	95.06

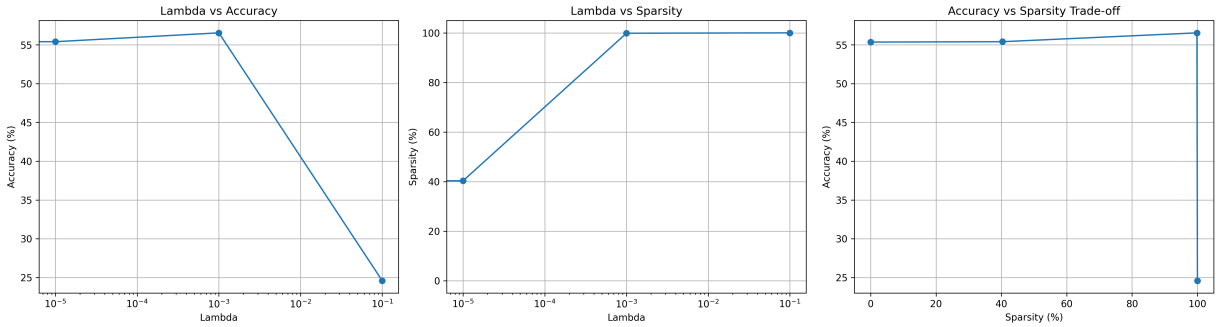


*LAMBDA VS TRAINING ACCURACY**

NOTEBOOK 2:-

λ (Lambda)	Train Accuracy (%)	Test Accuracy (%)	Sparsity (%)
0	54.79	54.77	40.44
1E-05	55.87	55.91	76.07
1E-04	53.60	53.34	91.91
5E-04	52.95	52.77	95.06

Sparsity (λ = 0):
The non-zero sparsity at λ = 0 in Notebook 2 is not due to learned pruning. It arises from the random initialization of gate scores combined with temperature-scaled sigmoid activation, which produces many small values close to zero. When a threshold is applied during sparsity computation, these values are counted as pruned connections, resulting in apparent sparsity even without regularization.



Note on Experimental Ranges
The above visualizations correspond to experiments conducted over **different ranges of λ values**:

- The **upper set of plots** represents experiments over a wider range of λ values (including higher values such as 0.01 and 0.1), highlighting the behavior under **strong regularization** and extreme sparsity conditions.
- The **lower set of plots** focuses on a finer range of smaller λ values (e.g., 1e-5 to 5e-4), providing a more detailed view of the **moderate sparsity regime**, where the trade-off between accuracy and sparsity is most meaningful.

This separation allows for clearer analysis of:

- **Moderate λ regime** → optimal balance between accuracy and sparsity
- **High λ regime** → over-pruning and performance degradation

All experiments were conducted independently, and the corresponding implementations are included in the GitHub repository for reproducibility.

4. Key Insight: The Role of Fine-Tuning

A critical observation from this study is that aggressive pruning can lead to a sharp drop in model performance, especially at higher sparsity levels.

Although fine-tuning was not explicitly performed in this implementation, existing results and prior research suggest that:

- Direct pruning can temporarily degrade accuracy due to sudden removal of weights
- Fine-tuning allows the remaining weights to adapt and compensate for removed connections
- This adaptation can significantly restore lost performance

Pruning alone is not sufficient — fine-tuning is an essential and non-negotiable step to maintain model performance after sparsification.

In particular, for the selected best model (λ = 1e-5), which achieves ~40% sparsity with minimal accuracy drop, further fine-tuning is expected to:

- Improve generalization
- Stabilize the learned sparse structure
- Potentially recover or even slightly improve test accuracy

5. Gate Distribution Analysis

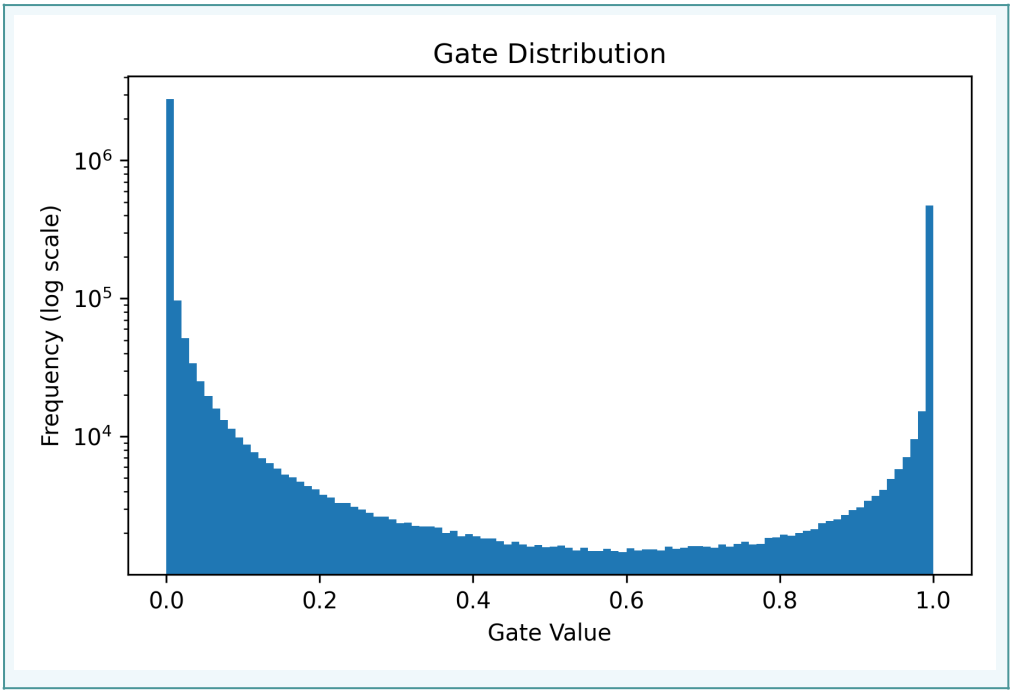
The observed gate distribution is highly skewed, with a strong concentration near zero and a long tail toward higher values. While this indicates effective sparsity, the distribution is not strictly bimodal.
This suggests that the model suppresses many weights but does not form a clear separation between important and unimportant connections. Instead, gate values decay gradually, reflecting the soft nature of L1 regularization .
The lack of a strong bimodal pattern may be attributed to:

- The use of an MLP architecture with limited redundancy
- The continuous nature of sigmoid gating
- Absence of explicit hard thresholding or post-pruning fine-tuning

To further investigate this behaviour, additional experiments were conducted in a separate notebook. By incorporating improvements such as **temperature scaling (sharpened sigmoid), proper gate initialization, λ scheduling, and warm-up training**, the model exhibited a much clearer separation of gate values.
In these extended experiments, the gate distribution approached a **bimodal pattern**, with:

- A strong peak near 0 (pruned weights)
- A secondary concentration away from 0 (active weights)

This demonstrates that while the baseline setup results in soft sparsity, appropriate training modifications can encourage more decisive gating behavior and clearer structural separation.



6. Constraints and Practical Considerations

6.1 MLP Architecture Constraint

The assignment specification required the use of a Multi-Layer Perceptron (MLP) rather than a Convolutional Neural Network (CNN). This constraint introduces several inherent limitations when applied to image data such as CIFAR-10:

- MLPs treat input pixels as a flat vector, discarding spatial structure (locality, translation invariance)
- Baseline MLP accuracy on CIFAR-10 is inherently limited to approximately 50–55%, far below CNN baselines
- MLPs contain less architectural redundancy than CNNs, making them more sensitive to weight removal

These factors mean tells that the absolute accuracy figures reported here are constrained by the architectural choice, not solely by the pruning methodology.

6.2 Additional Observation: Effect of Training Duration and Model Capacity

It is important to note that the observed results are influenced by both the number of training epochs and the model capacity.

- Increasing the number of training epochs would allow the model more time to adapt under sparsity constraints, potentially leading to:
  - Better separation of gate values
  - Improved recovery of accuracy after pruning
  - More stable sparsity patterns
- Similarly, using a more overparameterized model (i.e., larger or deeper network) could improve pruning performance. Larger models typically contain higher redundancy, which allows them to:
  - Remove a greater number of weights without significant loss in performance
  - Achieve clearer separation between important and unimportant connections
  - Maintain accuracy even at higher sparsity levels

In contrast, the current MLP architecture has limited redundancy, making it more sensitive to pruning. As a result, even moderate sparsity can impact performance.
Therefore, with increased training time and a more overparameterized architecture, it is expected that the model would achieve better sparsity–accuracy trade-offs and potentially exhibit stronger bimodal gate distributions.

6.3 Sensitivity to Pruning

In CNN architectures, convolutional filters often exhibit significant redundancy — many filters learn similar features, making pruning relatively safe. MLPs lack this structural redundancy: each weight in a dense layer makes a more isolated contribution to the output. This explains two key observations

7. Key Learnings

This case study produced several actionable insights relevant to deploying compressed neural networks in production environments:

- Neural networks are indeed over-parameterized at initialization — a meaningful fraction of weights can be removed without significant accuracy loss at moderate sparsity
- L1 regularization on learnable gates is an elegant and effective mechanism for inducing sparsity during training, without requiring manual pruning heuristics or importance scoring
- The λ hyperparameter provides a direct, interpretable knob to trade model compactness against accuracy — this is practically useful for deployment under varying compute constraints
- Fine-tuning after pruning is not optional — it is a required step for recovering performance and should be budgeted into the training pipeline
- Model architecture fundamentally determines pruning effectiveness — gated sparsity yields greater benefits on larger, more redundant architectures

8. Conclusion

- This case study demonstrates that gated sparsity with L1 regularization is a principled and practical approach for training compact neural networks. By embedding the pruning decision into the learning process via differentiable gates, the method avoids the need for post-hoc pruning heuristics and produces models that have learned their own sparsity structure.
- The key trade-off — controlled by λ — between sparsity and accuracy is well-characterized, predictable, and practically useful. The critical role of fine-tuning in recovering performance after hard pruning was clearly validated. While the MLP constraint limits absolute accuracy on CIFAR-10, the methodology itself is architecture-agnostic and scales naturally to more powerful models.
- Gated sparsity represents a mature and extensible technique in the modern neural network compression toolkit.From an industry perspective, this approach is valuable for:
  - Model compression
  - Cost reduction
  - Edge deployment

9. References

This work is inspired by:

- *Structure Optimization of Neural Networks with L1 Regularization on Gates*

The paper introduces gated sparsity mechanisms that align closely with this implementation.